Lecture 14

# Lines, planes and coordinates

**Uses.**

Summary 2: Standing fact E supplies trigonometric identities and the restricted inverse trigonometric functions.
Summary 12: Norms measure distances. Cauchy–Schwarz bounds dot products, which give angles and orthogonal decompositions with unique closest points.
Summary 13: The cross product gives an oriented normal. Independent plane directions give equivalent parametric and normal equations. Convention C fixes right-handed spatial axes. The vector triple-product identity has its full component proof on Sheet 13.

**Objectives.** *A line and a plane can be described by parameters or by scalar equations. We classify their intersections and compute angles and distances using normal components. Coordinate systems adapted to an axis then describe the same points by lengths and angles. Their parameter ranges must record the places where the description is not unique.*

By the end of this lecture you will be able to:

- Classify line–plane and plane–plane intersections with nonzero directions and normals.
- Prove and apply point–plane distance and line–plane angle formulas.
- Convert Cartesian, polar, cylindrical and spherical coordinates with $\theta$ polar and $\varphi$ azimuth.
- State coordinate ranges and identify the origin, axis and angular seam cases.

## 14.1 Lines and planes

A *line* through a point with position $\boldsymbol{p} \in \mathbb{R}^n$, in nonzero direction $\boldsymbol{d} \in \mathbb{R}^n$, is the set

$$L = \{\boldsymbol{p} + t\boldsymbol{d} : t \in \mathbb{R}\}.$$

We use $n = 2$ or $n = 3$ for plane or spatial geometry. The parameter of each point is unique: if $\boldsymbol{p} + t_1\boldsymbol{d} = \boldsymbol{p} + t_2\boldsymbol{d}$, then $(t_1 - t_2)\boldsymbol{d} = \boldsymbol{0}$. A nonzero component of $\boldsymbol{d}$ gives $t_1 = t_2$. A zero direction would describe one point, not a line.

For instance, the line with $\boldsymbol{p} = (2, 0, 0)$ and $\boldsymbol{d} = (1, 1, 2)$ has equations

$$(x, y, z) = (2 + t, t, 2t), \qquad t \in \mathbb{R}. \tag{14.1}$$

The point $(5, 3, 6)$ lies on it because all three equations hold at $t = 3$. The point $(5, 3, 7)$ does not, since its first two coordinates force $t = 3$ while its third coordinate disagrees. A point test must use a common parameter in every coordinate.

Recall from Lecture 13 that a plane in $\mathbb{R}^3$ has a *normal equation*

$$\Pi = \{\boldsymbol{x} \in \mathbb{R}^3 : \boldsymbol{n} \cdot \boldsymbol{x} = c\}, \quad \boldsymbol{n} \in \mathbb{R}^3 \setminus \{\boldsymbol{0}\}, \quad c \in \mathbb{R}.$$

The normal is orthogonal to every displacement within the plane. Equivalently, the plane has parameters $\boldsymbol{x} = \boldsymbol{a} + s\boldsymbol{u} + t\boldsymbol{v}$, with independent directions $\boldsymbol{u}$ and $\boldsymbol{v}$ satisfying $\boldsymbol{n} \cdot \boldsymbol{u} = \boldsymbol{n} \cdot \boldsymbol{v} = 0$, and any point $\boldsymbol{a} \in \Pi$. The equivalence and existence for every nonzero normal were proved in § 13.4. If all coefficients of a proposed normal equation are zero, the equation $0 = c$ instead describes all space when $c = 0$, or the empty set when $c \neq 0$.

## 14.2 Intersections, angles and distance

For a line $\boldsymbol{x} = \boldsymbol{p} + t\boldsymbol{d}$ and plane $\boldsymbol{n} \cdot \boldsymbol{x} = c$, substitution gives one scalar equation:

$$\boldsymbol{n} \cdot \boldsymbol{p} + t(\boldsymbol{n} \cdot \boldsymbol{d}) = c.$$

If $\boldsymbol{n} \cdot \boldsymbol{d} \neq 0$, there is exactly one intersection parameter,

$$t = \frac{c - \boldsymbol{n} \cdot \boldsymbol{p}}{\boldsymbol{n} \cdot \boldsymbol{d}}. \tag{14.2}$$

If $\boldsymbol{n} \cdot \boldsymbol{d} = 0$, every line point has the same normal component. The whole line lies in the plane when $\boldsymbol{n} \cdot \boldsymbol{p} = c$. Otherwise there is no intersection. A line direction orthogonal to the plane's normal is called *parallel to the plane*; this includes a line contained in the plane.

For a nonzero vector $\boldsymbol{u}$, its smaller angle with the unoriented line of direction $\boldsymbol{d}$ is

$$\beta = \arccos \frac{|\boldsymbol{u} \cdot \boldsymbol{d}|}{\|\boldsymbol{u}\| \, \|\boldsymbol{d}\|} \in [0, \pi/2].$$

Reversing either direction leaves this value unchanged. The *angle between a line and a plane* is the smallest of these angles as $\boldsymbol{u}$ varies over the nonzero directions in the plane. Split $\boldsymbol{d}$ into its part along the unit normal $\widehat{\boldsymbol{n}} = \boldsymbol{n}/\|\boldsymbol{n}\|$ and its part in the plane. The normal part has length $|\boldsymbol{d} \cdot \boldsymbol{n}|/\|\boldsymbol{n}\|$. The resulting right triangle therefore gives the angle formula

$$\alpha = \arcsin \frac{|\boldsymbol{d} \cdot \boldsymbol{n}|}{\|\boldsymbol{d}\| \, \|\boldsymbol{n}\|}. \tag{14.3}$$

For completeness, when the in-plane part is nonzero, its unit direction maximises the absolute dot product of $\boldsymbol{d}$ with in-plane unit directions by Cauchy–Schwarz. It therefore minimises the smaller angle with the line. When the in-plane part is zero, every such direction is orthogonal to $\boldsymbol{d}$, giving $\alpha = \pi/2$. Thus the right triangle gives the specified smaller angle in every case. Cauchy–Schwarz puts the displayed ratio in $[0, 1]$. Reversing either direction does not change this angle. A parallel line gives $\alpha = 0$.

**Theorem 14.1** (Point–plane distance). *Let* $\Pi$ *have equation* $\boldsymbol{n} \cdot \boldsymbol{x} = c$ *with* $\boldsymbol{n} \neq \boldsymbol{0}$. *For a point with position* $\boldsymbol{q}$, *its unique closest point on* $\Pi$ *is*

$$\boldsymbol{h} = \boldsymbol{q} - \frac{\boldsymbol{n} \cdot \boldsymbol{q} - c}{\|\boldsymbol{n}\|^2} \boldsymbol{n}.$$

*The distance to the plane is*

$$\operatorname{dist}(\boldsymbol{q}, \Pi) = \frac{|\boldsymbol{n} \cdot \boldsymbol{q} - c|}{\|\boldsymbol{n}\|}.$$

*Proof.* Taking a dot product with $\boldsymbol{n}$ gives $\boldsymbol{n} \cdot \boldsymbol{h} = \boldsymbol{n} \cdot \boldsymbol{q} - (\boldsymbol{n} \cdot \boldsymbol{q} - c) = c$, so $\boldsymbol{h}$ lies on the plane. For any other plane point $\boldsymbol{x}$, the displacement $\boldsymbol{x} - \boldsymbol{h}$ is orthogonal to $\boldsymbol{n}$. The displacement $\boldsymbol{q} - \boldsymbol{h}$ is a scalar multiple of $\boldsymbol{n}$. The two displacements are therefore orthogonal, and Pythagoras gives

$$\|\boldsymbol{q} - \boldsymbol{x}\|^2 = \|\boldsymbol{q} - \boldsymbol{h}\|^2 + \|\boldsymbol{x} - \boldsymbol{h}\|^2 .$$

The last term is nonnegative and vanishes only at $\boldsymbol{x} = \boldsymbol{h}$. This proves minimality and uniqueness. Taking the norm of the displayed normal displacement gives the distance formula. It also covers $\boldsymbol{q} \in \Pi$, when the distance is zero and $\boldsymbol{h} = \boldsymbol{q}$. ∎

Keep the plane from Lecture 13, $\Pi : \; x - 2y + z = 5$, and the line (14.1). Figure 14.1 shows a section containing this line and the normal direction. The section preserves the distances used below.

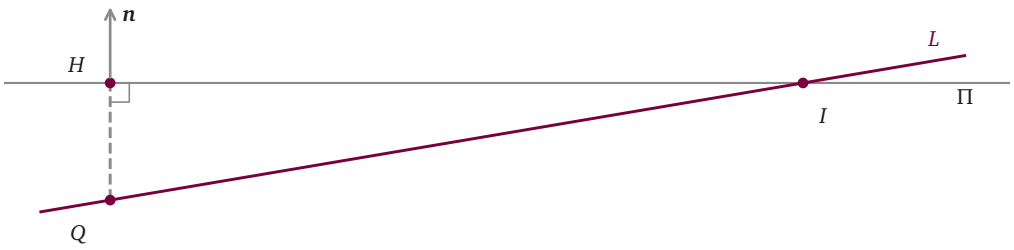



Figure 14.1: The line $L$ meets the plane at $I$. The point $H$ is the perpendicular foot of the starting point $Q$ on the plane. The drawing shows the two-dimensional section, not the spatial coordinate axes.

**Example 14.2** (One line, one plane). For this configuration, the normal is $\boldsymbol{n} = (1, -2, 1)$, the line starts at $Q = (2, 0, 0)$ and its direction is $\boldsymbol{d} = (1, 1, 2)$. The scalar data are

$$\boldsymbol{n} \cdot Q = 2, \qquad \boldsymbol{n} \cdot \boldsymbol{d} = 1, \qquad \|\boldsymbol{n}\| = \|\boldsymbol{d}\| = \sqrt{6}.$$

The intersection parameter is $(5 - 2)/1 = 3$, giving $I = (5, 3, 6)$. Substitution checks $5 - 2 \cdot 3 + 6 = 5$. The angle with the plane is $\arcsin(1/6)$. Using $\arccos(1/6)$ would instead give the smaller angle with the normal.

The perpendicular foot of $Q$ is

$$H = Q - \frac{2 - 5}{6}(1, -2, 1) = \left(\frac{5}{2}, -1, \frac{1}{2}\right).$$

It lies on the plane because $5/2+2+1/2 = 5$. The distance from $Q$ to the plane is $3/\sqrt{6} = \sqrt{6}/2$. The foot $H$ is not the intersection $I$. The line's direction is not the normal direction, so following the line to the plane does not give the shortest displacement.

## 14.3 Two planes

Consider equations $\boldsymbol{n}\cdot\boldsymbol{x}=c$ and $\boldsymbol{m}\cdot\boldsymbol{x}=d$, with $\boldsymbol{n},\boldsymbol{m}\in\mathbb{R}^3\setminus\{\boldsymbol{0}\}$ and $c,d\in\mathbb{R}$. If $\boldsymbol{m}=\lambda\boldsymbol{n}$ for a nonzero scalar $\lambda$, the planes coincide when $d=\lambda c$ and are disjoint otherwise. These conclusions follow by multiplying the first equation by $\lambda$.

If the normals are independent, put $\boldsymbol{w}=\boldsymbol{n}\times\boldsymbol{m}\neq\boldsymbol{0}$. Their intersection is a line in direction $\boldsymbol{w}$. First, a common point exists. One such position is

$$\boldsymbol{p}=\frac{c(\boldsymbol{m}\times\boldsymbol{w})+d(\boldsymbol{w}\times\boldsymbol{n})}{\|\boldsymbol{w}\|^2}.$$

Cyclic invariance of the triple product gives $\boldsymbol{n}\cdot(\boldsymbol{m}\times\boldsymbol{w})=\|\boldsymbol{w}\|^2$ and $\boldsymbol{m}\cdot(\boldsymbol{w}\times\boldsymbol{n})=\|\boldsymbol{w}\|^2$. The other two dot products vanish by cross-product orthogonality. Hence $\boldsymbol{n}\cdot\boldsymbol{p}=c$ and $\boldsymbol{m}\cdot\boldsymbol{p}=d$. Every point $\boldsymbol{p}+t\boldsymbol{w}$ satisfies both equations.

Conversely, subtract this point $\boldsymbol{p}$ from any common point and call the displacement $\boldsymbol{v}$. It obeys $\boldsymbol{n}\cdot\boldsymbol{v}=\boldsymbol{m}\cdot\boldsymbol{v}=0$. The vector identity proved by components on Sheet 13 states

$$(\boldsymbol{n}\times\boldsymbol{m})\times\boldsymbol{v}=\boldsymbol{m}(\boldsymbol{n}\cdot\boldsymbol{v})-\boldsymbol{n}(\boldsymbol{m}\cdot\boldsymbol{v})=\boldsymbol{0}.$$

The zero cross product makes $\boldsymbol{v}$ dependent on the nonzero $\boldsymbol{w}$. Thus $\boldsymbol{v}=t\boldsymbol{w}$, proving that the common set is exactly the claimed line. The explicit common-point formula proves existence; in a numerical problem, solving the two scalar equations can be shorter.

For example, intersect $x-2y+z=5$ with $2x+y-z=1$. Adding the equations gives $3x-y=6$. Choose $x=2+t$. Then $y=3t$ and the first equation gives $z=3+5t$. The intersection line is therefore

$$(x,y,z)=(2,0,3)+t(1,3,5),\qquad t\in\mathbb{R}.$$

The direction agrees with $(1,-2,1)\times(2,1,-1)=(1,3,5)$. The substitution derived every coordinate and allows every $t\in\mathbb{R}$, so it describes the whole intersection.

## 14.4 Polar and cylindrical coordinates

In the plane, *polar coordinates* describe a point by a radius $r$ and an azimuthal angle $\varphi$, measured counterclockwise from the positive first axis. The coordinate map is

$$(r,\varphi)\longmapsto(x,y)=(r\cos\varphi,r\sin\varphi),\qquad r\geq 0,\quad 0\leq\varphi<2\pi.$$

Here $r$ is distance from the origin, since $x^2+y^2=r^2$. For a nonzero point, $r=\sqrt{x^2+y^2}>0$ and there is exactly one $\varphi$ in the stated half-open interval with $\cos\varphi=x/r$ and $\sin\varphi=y/r$. To prove the angle choice from E, set $\gamma=\arccos(x/r)\in[0,\pi]$. Then $\sin\gamma=\sqrt{1-x^2/r^2}=|y|/r$. For $y\geq 0$, choose $\varphi=\gamma$. For $y<0$, choose $\varphi=2\pi-\gamma$; here $0<\gamma<\pi$, so the angle lies in $(\pi,2\pi)$. These choices have the required sine and cosine. Conversely, cosine on each half-circle determines one of these two forms by its restricted inverse and the identity $\cos(2\pi-\gamma)=\cos\gamma$. The sign of sine selects the form. When $y=0$, the cosine distinguishes $\varphi=0$ from $\varphi=\pi$. Thus the representation is unique. At the origin, $r=0$ and every permitted angle gives the same point. The polar angle is not defined there by the point alone.

If arbitrary real angles are allowed, adding $2\pi k$, with $k\in\mathbb{Z}$, gives the same point. The half-open range chooses one representative for nonzero points. Its cut along the positive

first axis is an *angular seam*: approaching that axis from above gives angles tending to 0, while approaching from below gives angles tending to $2\pi$. The coordinate angle jumps there even though the Cartesian points approach the same nonzero point.

For example, $(x, y) = (-1, \sqrt{3})$ has $r = 2$ and $\varphi = 2\pi/3$. The quotient $y/x = -\sqrt{3}$ alone would give principal arctangent $-\pi/3$, which is in the wrong quadrant. Both signs, or equivalently both sine and cosine conditions, determine the correct azimuth.

In space, *cylindrical coordinates* keep the third Cartesian coordinate:

$$(r, \varphi, z) \longmapsto (r\cos\varphi, r\sin\varphi, z), \quad r \geq 0, \quad 0 \leq \varphi < 2\pi, \quad z \in \mathbb{R}. \tag{14.4}$$

The radius now measures distance from the third axis. The inverse uses $r = \sqrt{x^2 + y^2}$ and the same planar angle. Off that axis the representation in these ranges is unique. On the axis, $r = 0$ and $\varphi$ is arbitrary. The same angular seam extends along the positive first half-plane from the axis.

## 14.5 Spherical coordinates and their exceptional points

In *spherical coordinates*, the radial distance from the spatial origin is denoted by $\rho$. This distinguishes it from the cylindrical distance $r$ from the third axis. The *polar angle* $\theta$ is measured from the positive third axis. The *azimuth* $\varphi$ is the same angle around that axis as before. This extends convention C with the course's angle names: $\theta$ is polar and $\varphi$ is azimuth. The coordinate map is

$$(x, y, z) = (\rho\sin\theta\cos\varphi, \rho\sin\theta\sin\varphi, \rho\cos\theta), \tag{14.5}$$

with ranges $\rho \geq 0$, $0 \leq \theta \leq \pi$ and $0 \leq \varphi < 2\pi$. The right triangle formed by the axis and the projected position gives

$$r = \rho\sin\theta, \qquad z = \rho\cos\theta, \qquad \rho^2 = r^2 + z^2 = x^2 + y^2 + z^2.$$

The two views in Fig. 14.2 separate the two angles.

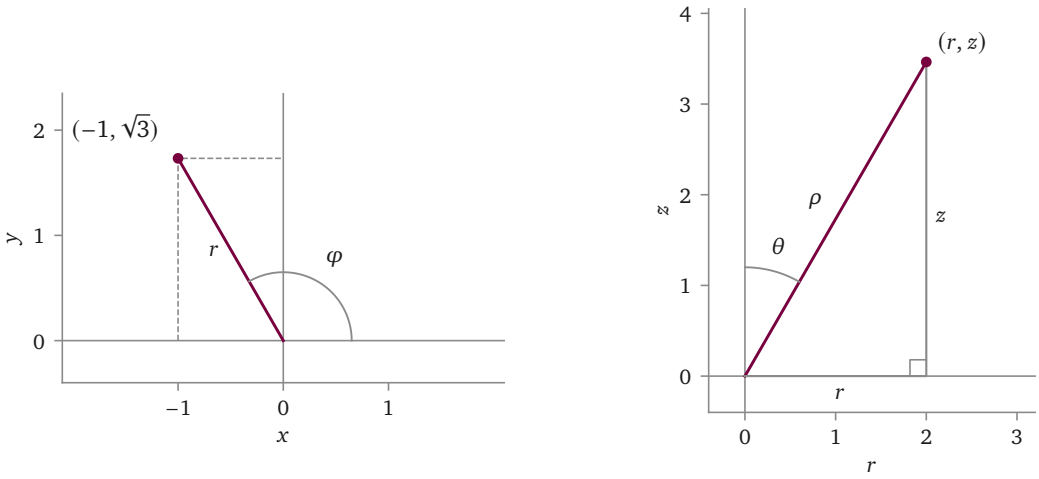



Figure 14.2: For the point $(-1, \sqrt{3}, 2\sqrt{3})$, the top view shows the azimuth $\varphi$. The meridian view with coordinates $(r, z)$ shows the polar angle $\theta$ measured from the positive third axis. The full radial distance is $\rho$, while the cylindrical radius is $r$.

For a point away from the origin, the inverse starts with

$$\rho = \sqrt{x^2 + y^2 + z^2} > 0, \qquad \theta = \arccos(z/\rho).$$

The ratio lies in $[-1, 1]$ since $|z| \le \rho$. Off the third axis, $r = \sqrt{x^2 + y^2} > 0$ and the polar inverse gives the unique $\varphi$ in $[0, 2\pi)$. Then $0 < \theta < \pi$ and $\rho \sin\theta = \sqrt{\rho^2 - z^2} = r$. Substituting these values in (14.5) recovers the original $x$, $y$ and $z$. This proves existence and uniqueness off the axis.

On the positive third axis, $\rho = z > 0$ and $\theta = 0$. On the negative third axis, $\rho = -z > 0$ and $\theta = \pi$. At either pole, the factor $\sin\theta$ is zero, so every permitted azimuth gives the same point. At the origin, $\rho = 0$ and both angles are arbitrary within their ranges. These are genuine non-uniqueness cases even after choosing a half-open azimuth interval. Away from the axis, the azimuth retains the same seam as cylindrical coordinates.

**Example 14.3** (One point in three coordinate systems)**.** Take the spatial point $(-1, \sqrt{3}, 2\sqrt{3})$. Its projection on the first two axes is the earlier polar example, so its cylindrical coordinates are $(2, 2\pi/3, 2\sqrt{3})$. The full distance and polar angle satisfy

$$\rho = \sqrt{1 + 3 + 12} = 4, \qquad \cos\theta = \frac{2\sqrt{3}}{4} = \frac{\sqrt{3}}{2}.$$

Thus its spherical coordinates are $(4, \pi/6, 2\pi/3)$. Substitution verifies

$$4 \cdot \tfrac{1}{2} \cdot (-\tfrac{1}{2}) = -1, \qquad 4 \cdot \tfrac{1}{2} \cdot \tfrac{\sqrt{3}}{2} = \sqrt{3}, \qquad 4 \cdot \tfrac{\sqrt{3}}{2} = 2\sqrt{3}.$$

The cylindrical and spherical radial coordinates are different lengths for this point.

**Example 14.4** (From a solid to spherical ranges)**.** Consider the part of the ball $x^2 + y^2 + z^2 \le 4$ that lies on or above the cone $z = \sqrt{x^2 + y^2}$. The meridian section in Figure 14.3 shows the distance $\rho$ and polar angle $\theta$; rotating this section about the $z$-axis gives the solid. For $\rho > 0$ the ball condition is $\rho \le 2$. The cone condition becomes $\rho\cos\theta \ge \rho\sin\theta$, hence $\cos\theta \ge \sin\theta$.

On $0 \le \theta \le \pi$ this is equivalent to $0 \le \theta \le \pi/4$: the inequality fails for $\theta \ge \pi/2$, and tangent is increasing on $[0, \pi/2)$. Thus one description is

$$0 \le \rho \le 2, \qquad 0 \le \theta \le \pi/4, \qquad 0 \le \varphi < 2\pi.$$

Conversely, substituting these ranges gives $x^2 + y^2 + z^2 = \rho^2 \le 4$ and $z = \rho\cos\theta \ge \rho\sin\theta = \sqrt{x^2 + y^2}$. The positive axis is included at $\theta = 0$, with arbitrary azimuth, and the origin is included at $\rho = 0$. The half-open azimuth range prevents an ordinary off-axis point from being counted twice at the seam.

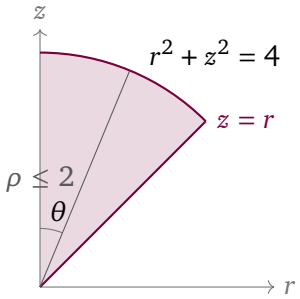



Figure 14.3: A meridian section with $r = \sqrt{x^2 + y^2} \ge 0$. The shaded sector is the part of the ball on or above the cone; its polar angles run from 0 to $\pi/4$.

## 14.6 An ellipse in polar coordinates

A *polar curve* prescribes a radius as a function of the azimuth. Consider

$$r = \frac{p}{1 + \varepsilon\cos\varphi}, \qquad p > 0, \quad 0 \le \varepsilon < 1, \quad 0 \le \varphi < 2\pi. \tag{14.6}$$

The denominator is at least $1 - \varepsilon > 0$, so the radius is always positive. To identify the curve, use $x = r\cos\varphi$ and $r^2 = x^2 + y^2$. Multiplication by the denominator gives $r + \varepsilon x = p$. Squaring $r = p - \varepsilon x$ and collecting terms gives

$$(1 - \varepsilon^2)x^2 + y^2 + 2\varepsilon p x - p^2 = 0.$$

Completing the square yields

$$(1 - \varepsilon^2)\left(x + \frac{\varepsilon p}{1 - \varepsilon^2}\right)^2 + y^2 = \frac{p^2}{1 - \varepsilon^2}.$$

An *ellipse* with centre $(h, 0)$ and positive semiaxes $a$ and $b$ has equation $(x-h)^2/a^2 + y^2/b^2 = 1$. The horizontal and vertical semiaxes are the distances from the centre to the intersections with the horizontal and vertical lines through it. Dividing by the positive right side therefore identifies the centre and semiaxes of our curve:

$$h = -\frac{\varepsilon p}{1 - \varepsilon^2}, \qquad a = \frac{p}{1 - \varepsilon^2}, \qquad b = \frac{p}{\sqrt{1 - \varepsilon^2}}.$$

Squaring could have added points, so the converse needs its sign check. Every point of this ellipse satisfies $x \le h + a = p/(1 + \varepsilon)$. Consequently

$$p - \varepsilon x \ge p - \frac{\varepsilon p}{1 + \varepsilon} = \frac{p}{1 + \varepsilon} > 0.$$

The squared equation therefore implies $\sqrt{x^2+y^2} = p - \varepsilon x$ with the required positive sign. The origin does not satisfy the equation since $p > 0$. Every ellipse point thus has a nonzero radius and a standard polar angle. Substituting $x = r\cos\varphi$ recovers (14.6). This proves equality of the two curves in both directions. When $\varepsilon = 0$, the centre is the origin and both semiaxes are $p$, so the curve is a circle. For $0 < \varepsilon < 1$, its centre is to the left of the polar origin.

Figure 14.4 shows the case $p = 2$ and $\varepsilon = 3/5$ together with its polar origin. For this

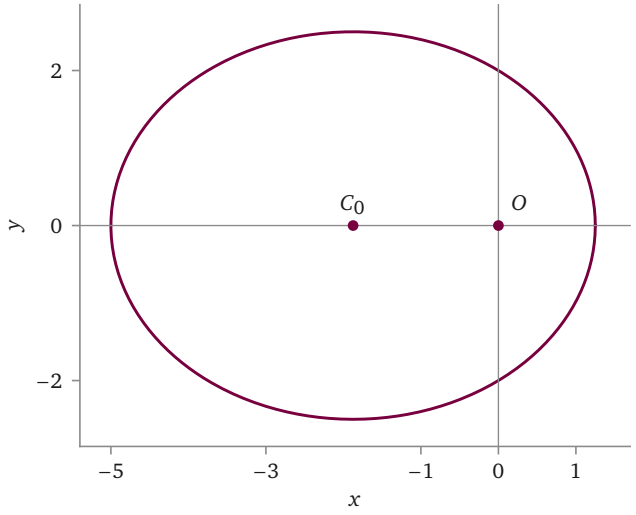



Figure 14.4: The curve $r = 2/(1 + (3/5)\cos\varphi)$ has centre $C_0 = (-15/8, 0)$. The polar origin $O$ differs from the centre.

choice of parameters, the formulas give centre $(-15/8, 0)$, horizontal semiaxis $25/8$ and vertical semiaxis $5/2$.

**Summary 14.**

- A line has a nonzero direction. Substituting it into a plane's normal equation gives one intersection, a contained line or no intersection, according to the scalar coefficient and constant.
- Equation (14.3) and Theorem 14.1: the line–plane angle uses an absolute normal component and arcsine. The closest plane point is obtained by removing the signed normal displacement.
- Independent plane normals give an intersection line in their cross-product direction. Proportional normals give coincident or disjoint planes.
- Equations (14.4) and (14.5): cylindrical $r$ is distance from the third axis, while spherical $\rho$ is distance from the origin. The angle $\theta$ is polar and $\varphi$ is azimuth.
- A half-open azimuth range removes periodic duplication off the axis. It leaves an angular seam. The origin and axis retain the specified non-uniqueness of angles.

The polar curve (14.6) is an ellipse, with a circle as its $\varepsilon = 0$ case. The sign check after squaring is essential. Sheet 14 treats parameter cases, skew lines and coordinate regions, and derives a polar equation for a centred ellipse. Lecture 15 gives a multiplication to the coordinate plane to construct complex numbers and describe rotations.

## References

J. Stewart, *Calculus: Early Transcendentals*, 8th ed., Cengage, 2016, §§10.3,12.5 and 15.7–15.8. K. F. Riley, M. P. Hobson and S. J. Bence, *Mathematical Methods for Physics and Engineering*, 3rd ed., Cambridge University Press, 2006, §7.7.